EXP NO :
17/09/26 SLIDING WINDOW PROTOCOL

AIM :

Write a program to implement flow chart at data link layer using SLIDING WINDOW PROTOCOL. simulate the flow of frames from one node to another.

CODE :

sender :

```
import time
import os
window_size = int(input("Enter window size:"))
message = input("Enter the message :")
frames[]
for i, ch in enumerate(message, start = 1):
    frames.append((i, ch))
sender_buffer = "sender_buffer.txt"
receiver_buffer = "Receiver_buffer.txt"
start = 0
while start < len(frames):
    window = frames[start : start + window_size]
    print("\nsender")
    with open(sender_buffer, "w") as file:
        for frame_no, data in window:
            file.write(f"{frame_no}, {data}\n")
            print("Frame", frame_no, ":", data)
    input("\n press enter to continue.")
    if not os.path.exists(receiver_buffer):
        print("Receiver.txt not found")
        continue
    print("Received :", responses)
    if response.startswith("ACK :"):
        ack_no = int(response.split(":"))
```

```
                [1])
            expected_ack = window[-1][0]
            if ack_no == expected_acx:
                print("ACK", ack_no, "received")
                print("window moved")
                print("window moved")
                start = start.window_size
            else:
                print("Wrong ACK")
                print("Expected :", expected_ack)
                print("Received", ack_no)
                print("Resending frames")
        elif response.startswith("NACK:"):
            nack_no = int(response.split
                          (":")[1])
            print("NACK", nack_no,)
            start = nack_no - 1
        else:
            print("Invalid response")
        time.sleep(1)
    print("\n All frames are sent")
```

Receiver:

```
import os
sender_buffer = "sender_Buffer.txt"
received_buffer = "receiver_Buffer.txt"
expected_frame = int(input("enter expected
                           frame no:"))
print("\n Receiver)
if not os.path.exists(sender_buffer):
    print("sender_Buffer.txt not found")
else:
    with open(sender_buffer, "r") as file:
        lines = file.readings()
        error = false
```

```
        last_Frame = expected_frame - 1
        print ("Frames received")
        for line in lines:
            line = line.strip()
            if line == "":
            continue
        parts = line.split(",")
        frame_no = int(parts[0])
        data = parts[1]
        print("Frame", frame_no, ":", data)
        last_Frame = frame_no
        expected_frame += 1
    else:
        print("Error in frames", frame_no)
        print("Expected frames:", expected_Frame)
        error = True
        with open(receiver_buffer, "w") as file,
            file.write("NACK:{expected_frame}")
        break
if not error:
    with open(receiver_buffer, "w") as file:
        file.write(f"ACK:{last_frame}")
    print("ACK", last_frame, "sent")
```

OUTPUT:

```
sender(window1):
    Enter window size: 3
    Enter the message: HELLO
Sender
    frame 1: H
    frame 2: E
    frame 3: L
```

Frames written to sender_Buffer.txt

Received : ACK : 3

ACK 3 received

window moved.

sender

Frame 4 : L

frame 5 : O

Frames written to sender_Buffer.txt

Received : ACK : 5

ACK 5 received

window moved.

All frames sent sucessfully.

Receiver (window 2):

Expected frame no: 1

Receiver

Frame 1 : H

received correctly

Frame 2 : E

received correctly

Frame 3 : L

Ack 3 send

Expected frame no: 4

Frame 4:

received correctly

Frame 5 :

received correctly.